PQC DAILY STUDY SERIES

PQC's Impact on Blockchain and Cryptocurrencies

Day 36 | Category: PQC Adoption & Industry Status Date: 2026-10-21 | **DAILY MONITORING**

PQC's Impact on Blockchain and Cryptocurrencies

As quantum computers become more powerful, the cryptographic foundations of blockchain and cryptocurrencies face an existential threat. This study explores the vulnerabilities and the ongoing transition to quantum-resistant solutions.

The Quantum Threat to Blockchain

Blockchain's security relies heavily on classical cryptographic algorithms, primarily the Elliptic Curve Digital Signature Algorithm (ECDSA) for digital signatures. These signatures are used to authorize transactions and prove ownership of assets. Shor's algorithm, which can run on a sufficiently powerful quantum computer, can break ECDSA, allowing malicious actors to forge signatures and steal funds.

> **Key Vulnerability:** Address reuse. When a cryptocurrency address is used multiple times, its public key is exposed, making it a target for quantum attacks.

Transitioning to Quantum-Resistant Blockchains

The blockchain community is actively researching and developing quantum-resistant solutions. The primary approach is to replace ECDSA with PQC digital signature algorithms like ML-DSA and SLH-DSA.

Key Challenges in the Transition

- **Signature Size:** PQC signature algorithms often have larger signature sizes compared to ECDSA, which can increase transaction size and fees.
- **Verification Time:** PQC algorithms may have different performance characteristics, impacting transaction verification times and overall network throughput.
- **Forking:** Upgrading a blockchain to be quantum-resistant often requires a hard fork, which can be a contentious and disruptive process.

ECDSA vs. PQC Signatures (Illustrative)

Algorithm	Key Size (bytes)	Signature Size (bytes)	Quantum Resistance

Algorithm	Key Size (bytes)	Signature Size (bytes)	Quantum Resistance
ECDSA (secp256k1)	32 (private), 65 (uncompressed public)	64-72	No
ML-DSA-44	~1.3KB (private), ~1.3KB (public)	~2.4KB	Yes
SLH-DSA-128s	~32 (private), ~16 (public)	~7.8KB	Yes

Daily Quiz

1. What is the primary quantum threat to cryptocurrencies like Bitcoin?
2. What is one of the main challenges of transitioning a blockchain to use PQC algorithms?
3. Which NIST-standardized PQC algorithm is a candidate for replacing ECDSA in blockchains?
